Foundation Models for Water Surface Mapping: A Study with Copernicus-FM

Author: Farshad Farahnakian [1]

Supervised by: Prof. Jukka Heikkonen[1] and Dr. Fahimeh Farahnakian[2]
1 Department of Computing, University of Turku, Finland
2 Geological Survey of Finland (GTK)

Abstract:

Background & gap. Surface-water mapping underpins flood response, water-security planning, and wetland conservation, yet conventional rule-based indices and task-specific deep networks struggle to generalize across regions and seasons or demand extensive labels. Recent Earth-observation foundation models (FMs) promise transferable representations, but their robustness beyond their pretraining horizon remains under-documented. In particular, the out-of-scope performance of Copernicus-FM on application-driven datasets for water mapping has not been empirically established.

Method. We evaluate a frozen Copernicus-FM backbone paired with a lightweight fully-convolutional decoder for binary water-surface segmentation on the Water Bodies dataset. We quantify label efficiency by training with {10, 20, 30, 50, 100}% of the training tiles while keeping validation/test splits fixed. Performance is reported with Overall Accuracy (OA), Intersection-over-Union (IoU), and Dice.

Results. The model attains Test OA 0.864, IoU 0.749, and Dice 0.855 at full supervision, with strong performance already at 10% labels (IoU 0.683, Dice 0.809). Gains concentrate between 10–30% labels, after which returns saturate; at 30%, the system recovers ~98% of fully-supervised OA/Dice and ~97.5% of IoU. Confusion matrices show decreasing false negatives (12.2%→9.0%) and false positives (5.4%→4.5%), indicating cleaner shoreline and thin-channel delineation. Model size (~13 MB trainable) and inference latency (sub-ms per image) are constant across fractions.

Discussion & implications. These findings demonstrate that Copernicus-FM generalizes effectively beyond its pretraining benchmark to an external, and delivers high-quality water masks with scarce labels, directly addressing concerns about domain specificity. The clear label-efficiency knee (20–30%) provides a practical operating point for rapid, low-cost deployment in resource-constrained regions.

Algorithmic and Computational Intelligence Research Group

UNIVERSITY OF TURKU

ThinkingEarth

EUSPA

GitHub code:

https://github.com/FarFar-Tech/Water-Surface-Mapping-using-Satellite-Imagery

Introduction

Mapping surface water has long been a central application of Earth observation (EO). Clear and timely mapping of water areas is essential for flood warnings, planning for water security, protecting wetlands, modeling water systems, and developing groundwater potential adaptation strategies (see Figure 1). This is important at both local and global levels. Automated and semi-automated approaches typically rely on rule-based systems, machine learning (ML), or hybrids of the two. Rule-based methods set thresholds on spectral bands or compute multiband indices; ML methods learn decision boundaries from data.

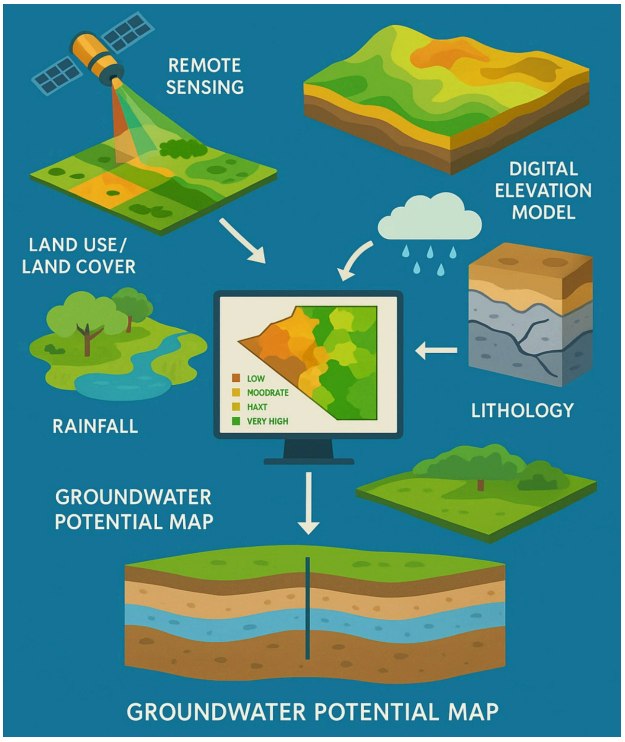

Figure 1: This is how water surface mapping can be used in a groundwater potential map.

A widely used rule-based approach for multispectral imagery (e.g., Landsat) involves computing two-band water indices, such as the Normalized Difference Water Index (NDWI) or its modified variant, the Modified Normalized Difference Water Index (MNDWI). These indices leverage the distinct reflectance of water in visible and infrared wavelengths to enhance water features, followed by thresholding to obtain binary masks.

However, index-based methods suffer from three practical limitations: (i) the “best” threshold varies across geography, season, sensor, and illumination, undermining global applicability; (ii) even optimally tuned thresholds often confuse water with snow, ice, and terrain shadows; and (iii) high-quality rule-based pipelines frequently depend on complex, region-specific heuristics and ancillary data (DEMs, inventories), which reduces scalability and portability.

To overcome these constraints, the community has adopted deep learning (DL) models, including Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). While CNNs capture local texture and edges effectively, they can miss long-range dependencies.

ViTs better model global context through self-attention, but are computationally intensive and typically require substantial labeled data—both of which limit their practical deployment at continental or global scales.

Recent progress in Foundation Models (FMs) for EO suggests a promising alternative. Pretrained on massive and diverse satellite corpora, FMs can produce rich, transferable representations that adapt rapidly to new tasks with modest labeled data. In principle, FMs can (i) reduce annotation burden, (ii) complement task-specific DL models, and (iii) improve robustness under domain shift. Yet, two critical gaps persist: (1) performance of specific EO FMs—such as Copernicus-FM—has not been empirically established on datasets and regions outside their pretraining scope for environmental monitoring tasks like water and flood mapping; and (2) the label-efficiency profile of these models (how much annotation is truly needed) remains under-quantified for operational settings.

This study addresses both gaps. We investigate whether Copernicus-FM, pretrained on large-scale, heterogeneous EO data, can deliver accurate surface-water maps when fine-tuned with limited task-specific labels. Specifically, we train a lightweight decoder on top of Copernicus-FM using {10%, 20%, 30%, 50%, 100%} of the available training tiles, while maintaining the validation and test sets. We conduct experiments on three water-surface datasets spanning diverse conditions to probe generalization beyond Copernicus-FM's original training distribution.

Research Questions (RQs)

- RQ1: How can FMs empower the regions that need it most by making water surface mapping more affordable and accessible?
- RQ2: How effectively can CopernicusFM perform when faced with a dataset beyond its pretraining horizon?

The Contributions of this study are as follows:

- Robust Water Mapping Beyond Pretraining Scope. We provide the first, to our knowledge, systematic evaluation of Copernicus-FM for surface-water segmentation across two external datasets, demonstrating how FM features transfer to environmental monitoring and flood-related use cases.
- Label-Efficiency Curves with Fixed Validation/Test. We quantify performance across five label budgets (1–100%) using a frozen-backbone + lightweight decoder protocol, showing where returns saturate and what minimum annotation is needed for operational quality.
- Transparent cost accounting. Beyond accuracy, we report parameter count, model size (MB), training time (s), and inference latency (ms/image) for every fraction. These measurements make the trade-offs between accuracy and cost explicit and actionable for agencies with limited computing or budgets.
- A scalable solution that can be adapted for analyzing additional inputs and for each region targeted by remote sensing researchers and projects.

Methodology

This section describes how we utilize a frozen Copernicus-FM encoder with a light segmentation decoder and label-efficiency fine-tuning to address two research questions.

Dataset: We utilize the public Satellite Images of Water Bodies dataset from Kaggle, which was originally compiled from Sentinel-2 imagery and is distributed as 2,841 RGB images paired with 2,841 binary masks (where white represents water and black represents non-water). The collection spans multiple continents, climate zones, and seasons, and includes a broad range of water types (rivers, lakes, ponds, reservoirs, and coastal segments). Masks were produced using a water-index–based procedure (NDWI/MNDWI-style) and manually verified for training use. Figure 2 shows different samples of this data, along with their corresponding masks. The link of the data:

https://www.kaggle.com/datasets/franciscoescobar/satellite-images-of-water-bodies

Why this dataset? It offers: (i) diverse geography and illumination conditions; (ii) varied hydrological forms that stress different spatial scales and backgrounds; (iii) seasonal variability that tests robustness under domain shift; and (iv) curated annotations suitable for supervised segmentation. These characteristics align with our study's goals of evaluating generalization beyond the pretraining scope and label efficiency for water mapping.

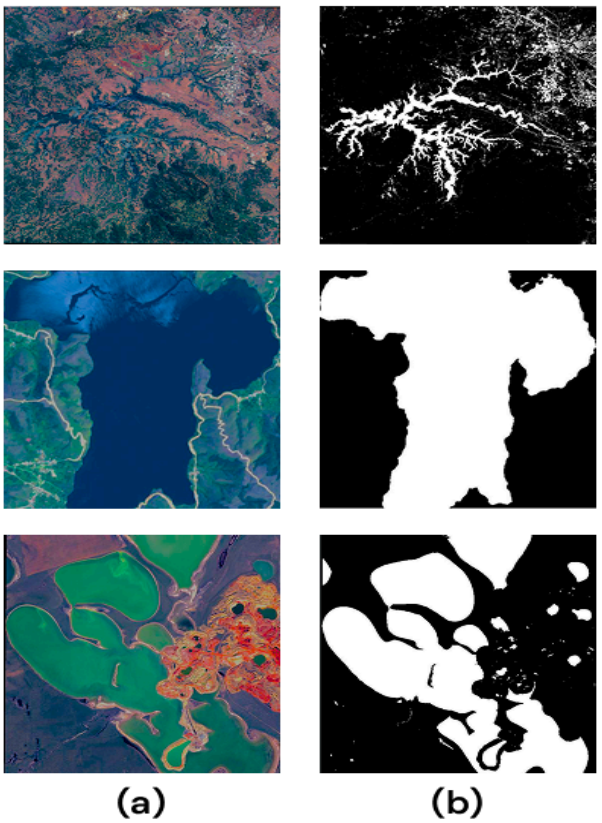

Figure 2: Samples of the Water Bodies dataset: (a) RGB image, (b) mask.

Preprocessing: images and masks are converted to tensors, then letterboxed to a square—we resize the shortest side and center-crop/pad to 256×256 so aspect ratio semantics are preserved; masks are resized with nearest-neighbor and remain strictly binary (water vs. background). Inputs are per-channel normalized using mean/std estimated on the training split only. During training, we apply lightweight augmentations (random flips and 90° rotations) to improve robustness while preserving shoreline geometry. We use fixed validation and test sets shared across all label-budget experiments to ensure comparability, with no geographic leakage between splits.

Our pipeline adapts the Copernicus-FM Earth-observation foundation model for binary water-surface segmentation from Sentinel-2 imagery. As shown in Figure 3, a frozen Copernicus-FM encoder produces dense feature maps that a lightweight decoder converts into a water mask. The encoder remains fixed during training—only the small decoder is optimized—which (i) preserves Copernicus-FM's rich Copernicus-scale priors and (ii) makes training fast and label-efficient.

Encoder:

- Backbone: "copernicusfm_based" initialized with CopernicusFM_ViT weights (called from TochGeo python library).
- Inputs: 3-band images (RGB) resized to 256×256 and normalized.
- Spectral contract: We specify wavelengths in nanometers (≈[480, 560, 650]) and simple fixed bandwidths (≈[50, 50, 50] nm). This tells the encoder which parts of the pretrained spectral space the RGB channels correspond to—no geographic or temporal metadata is required.
- Tokenization: A ViT-style patch size of 16×16 yields $(256/16)^2$ = 16×16 tokens. The encoder returns a token matrix [B, N, D], which we reshape into a spatial feature map [B, D, S, S] with S=16 and D≈768.
- Frozen weights: All Copernicus-FM parameters are kept frozen. This isolates the representation quality learned during pretraining, allowing us to measure how much labeled data the task head requires—central to RQ2.

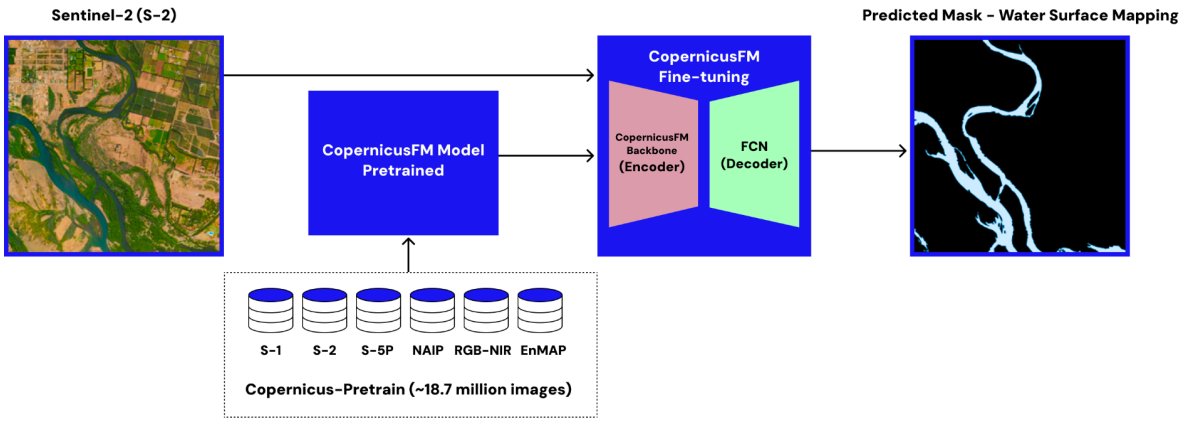

Figure 3: The overview of the proposed pipeline leveraging Copernicus-FM.

Decoder (small, stable, and fast):

We attach a lightweight fully convolutional head to the frozen Copernicus-FM encoder to produce binary water masks. The head consists of two 3×3 convolutional blocks that progressively reduce the feature dimensionality (from the encoder's high-dimensional map to a compact representation), each followed by Group Normalization, ReLU activation, and dropout (p=0.2) for regularization. A final 1×1 convolution projects this representation to two logits (water vs. background). The logits are then upsampled with bilinear interpolation to the input image resolution (no learnable upsampling), ensuring alignment with pixel-level labels. This design is intentionally parameter-efficient (~3.3M trainable parameters) and memory-light, which (i) stabilizes optimization with small batch sizes and scarce labels, (ii) shortens training time, and (iii) preserves the strong priors learned by Copernicus-FM during pretraining.

Fine-tuning protocol for this research is as follows:

- Only the decoder parameters; the Copernicus-FM backbone remains frozen.
- Task: binary semantic segmentation (water vs. non-water).
- Splits: A fixed **train/val/test** partition is created once (e.g., 80/10/10) across all label-budget experiments to ensure fair, comparable evaluation (see Figure 4)—only the number of **labeled training images** changes.
- Label-efficiency study: For each budget, we train the same decoder architecture on the corresponding fraction of training tiles and report performance on the unchanged validation/test sets.

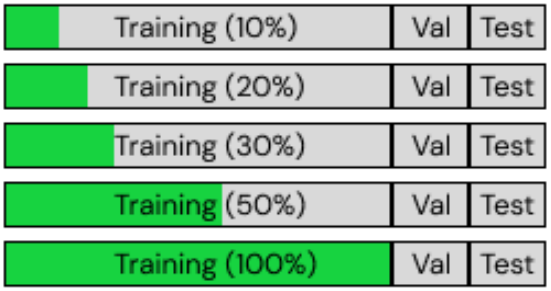

Figure 4: Fine-tuning strategy

Evaluation Metrics: We evaluate binary water-surface segmentation with three standard, complementary metrics computed at the pixel level on the fixed test set: Overall Accuracy (OA), Intersection-over-Union (IoU), and Dice coefficient (F1-score). Pixels labeled as "ignore/void" (if any) are excluded from all denominators.

For the positive class water:

- **TP**: predicted water & labeled water
- **FP**: predicted water & labeled non-water
- **FN**: predicted non-water & labeled water
- **TN**: predicted non-water & labeled non-water

Overall Accuracy (OA) — fraction of correctly classified pixels:

$$\mathrm{OA} = \frac{TP + TN}{TP + FP + FN + TN}$$

Why OA? High-level correctness across both classes; it can be inflated by background dominance, so we pair it with class-focused metrics below.

Intersection-over-Union (IoU) — overlap quality for a class mask:

$$\mathrm{IoU}_{\mathrm{water}} = \frac{TP}{TP + FP + FN}$$

Why IoU? Stricter than accuracy; penalizes both over- and under-segmentation.

Dice coefficient (F1) — harmonic mean of precision and recall for water:

$$\mathrm{Dice}_{\mathrm{water}} = \frac{2TP}{2TP + FP + FN}$$

Why Dice? Correlate with operational needs, where boundary fidelity matters.

Results

Experimental setup: We train a frozen Copernicus-FM encoder with a small decoder (≈ 3.32M trainable parameters, 13.27 MB on disk) for binary water/background segmentation on the waterbody dataset (non-georeferenced RGB). To study label efficiency, we keep the validation and test sets fixed and vary the fraction of labeled training tiles: {10, 20, 30, 50, 100}% of the total training set. For each fraction, we train the same decoder for 100 epochs, with a batch size of 16, and evaluate using evaluation metrics (OA, mIoU, Dice).

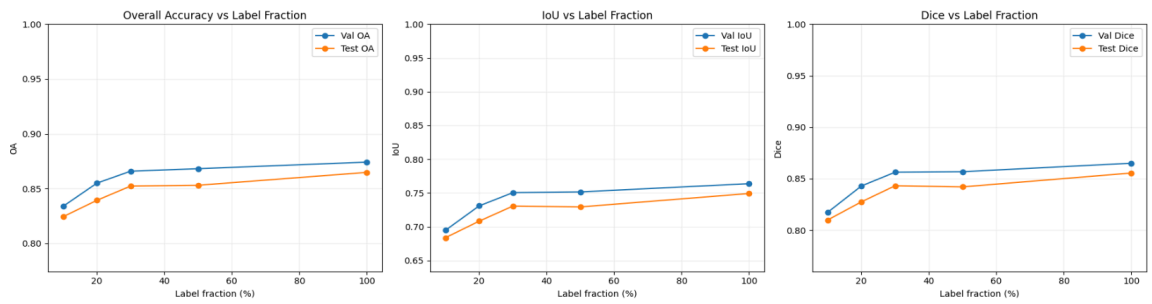

Figure 5: Fine-tuning strategy

Quantitative results: Figure 5 (label-efficiency curves) and Table 1 summarize the results obtained for each fraction. All three metrics show monotonic improvement as the label

fraction increases. Throughout all fractions, the gap between validation and test results remains small, typically around 1-2 percentage points, with the validation and test curves closely aligned (see Figure 5). This indicates that the frozen-backbone strategy paired with a compact head facilitates stable generalization without evident signs of overfitting, even as the volume of labels increases.

The most significant improvements in Table 1 are observed when transitioning from 10% to 30% labeled data, after which enhancements gradually plateau (notably between 30% and 50%) before modest gains resume at 100%. Even with just 10% of the labels, the model achieves a Test Dice of approximately 0.81 and a Test IoU of around 0.68, demonstrating strong performance under limited supervision.

Table 1. Results obtained for each fraction based on OA, IoU, and Dice metrics.

Fraction	Val_OA	Val_IoU	Val_Dice	Test_OA	Test_IoU	Test_Dice
10	0.83	0.694	0.817	0.824	0.683	0.809
20	0.85	0.730	0.842	0.839	0.708	0.827
30	0.86	0.750	0.856	0.852	0.730	0.843
50	0.86	0.751	0.856	0.853	0.729	0.841
100	0.87	0.763	0.864	0.864	0.749	0.855

Table 2. Model complexity across fractions.

Fraction	Number of training images	Total of parameters	Parameters trainable	Model size (MB)	Train time (s)	infer_ms_per_img
10	227	3319874	3319874	13.27	795.46	0.110
20	455	3319874	3319874	13.27	1107.42	0.903
30	682	3319874	3319874	13.27	1426.55	0.980
50	1136	3319874	3319874	13.27	2047.18	0.829
100	2273	3319874	3319874	13.27	3601.21	0.786

Table 2 shows that model capacity remains unaffected by the label budget: the decoder head comprises approximately 3.32 million parameters (around 13.27 MB), and as the CopernicusFM backbone is kept frozen, the count of trainable parameters does not change with additional training data. As anticipated, training duration increases roughly in proportion to the number of training images, ranging from approximately 800 seconds to 3600 seconds when scaling from 10% to 100% of the dataset. This reflects a rise in optimization steps rather than an increase in complexity. The inference latency per image consistently stays below a millisecond, fluctuating between 0.79 and 0.98 milliseconds, with these minor variations attributed to benchmarking noise such as I/O, caching, and batch effects.

Practically, accuracy improves with the addition of more labels, without affecting the model's size or runtime costs, thus maintaining a low-footprint and production-ready profile.

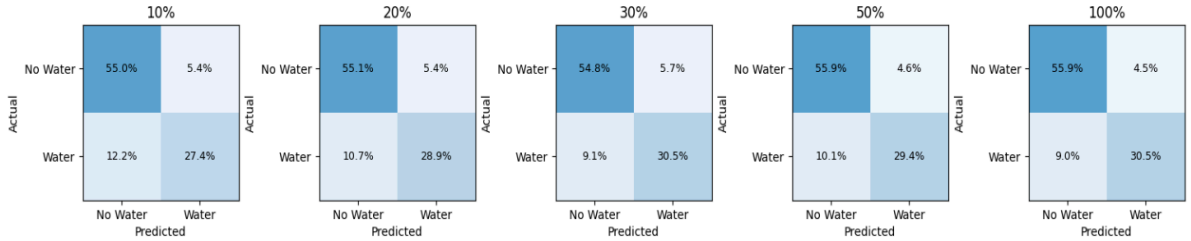

Figure 5: Confusion matrix based on different fractions.

The confusion matrix (Figure 6) reveals consistent gains in both recall and precision for the water class as supervision increases. False negatives (missed water) drop from 12.2% → 9.0%, while false positives (spurious water) decline from 5.4% → 4.5%; correspondingly, true positives rise from 27.4% → 30.5% and true negatives remain stable around 55–56%, reflecting the dataset's background dominance. These changes translate to water recall improving from ~69% (10%) to ~77% (100%) and water precision from ~84% to ~87%. Overall, the matrices show that additional labels primarily reduce missed water along narrow channels and shorelines while modestly curbing over-segmentation, yielding cleaner, more reliable masks.

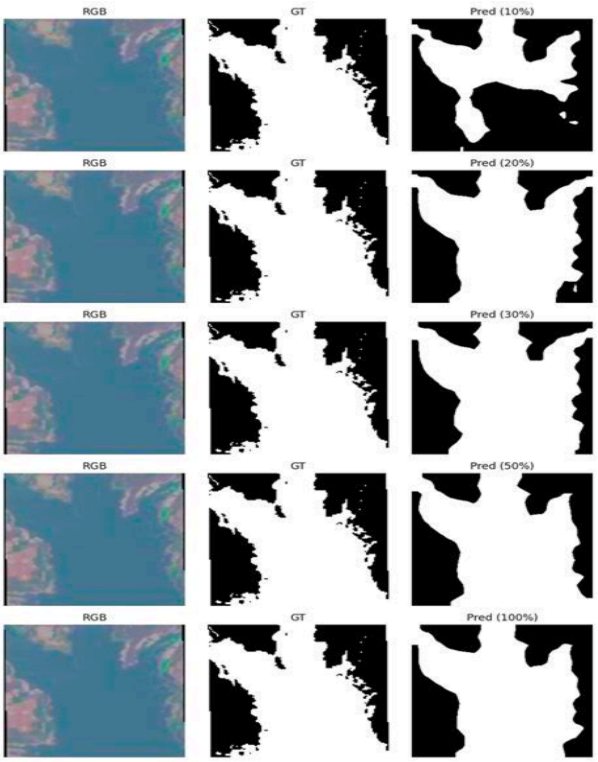

Figure 7: Sample predictions of water surface mapping based on each fraction.

Qualitative results: Figure 7 shows that predicted masks delineate water bodies cleanly, with typical errors concentrated along shorelines, small inlets, and areas with sunglint/sediment. Qualitative quality improves with label fraction, but the visible differences between 30% and 100% are subtle—consistent with the quantitative saturation.

Discussion

Comparison: To explore what traditional DL-based models work, we make a fair comparison; the UNet experiments were run on the same dataset and fixed validation/test splits, with the same preprocessing (letterbox-to-square resizing, binary masks, per-channel normalization on the training split) and evaluation metrics (OA/IoU/Dice). We trained UNet-ResNet34 (ImageNet) end-to-end using the same label-fraction protocol. Tables 3 and 4 show the obtained results.

Table 3. Results obtained for each fraction based on OA, IoU, and Dice metrics - UNet-ResNet34.

Fraction	Val_OA	Val_IoU	Val_Dice	Test_OA	Test_IoU	Test_Dice
10	0.71	0.305	0.366	0.71	0.329	0.394
20	0.78	0.285	0.354	0.77	0.299	0.373
30	0.80	0.356	0.413	0.80	0.384	0.445
50	0.77	0.320	0.389	0.76	0.338	0.409
100	0.80	0.355	0.4184	0.79	0.373	0.438

Table 4. Model complexity across fractions - UNet-ResNet34.

Fraction	Number of training images	Total of parameters	Parameters trainable	Model size (MB)	infer_ms_per_img
10	227	24436514	24436514	93.384	229.369
20	455	24436514	24436514	93.384	224.335
30	682	24436514	24436514	93.384	228.607
50	1136	24436514	24436514	93.384	223.533
100	2273	24436514	24436514	93.384	230.134

Answering the research question

RQ1: How can FMs empower the regions that need it most by making water-surface mapping more affordable and accessible?

Affordability via label efficiency: Across identical label budgets (10, 20, 30, 50, 100%), Copernicus-FM (frozen encoder + light decoder) consistently and substantially outperforms a strong UNet-ResNet34 (ImageNet) baseline. At 10% labels, Copernicus-FM reaches Test IoU ≈ 0.68 versus ≈ 0.33 for UNet (ΔIoU ~ +0.36). The advantage persists at every fraction and remains ~+0.35–0.41 IoU even at full supervision. This means 70–90% fewer labels can already push FM performance to a regime that the baseline cannot match, directly reducing the dominant cost in operational mapping—manual annotation.

Accessibility through lightweight, deployable models: The FM setup fine-tunes only a small decoder (≈ 3.32M params; ~13 MB on disk) while keeping the backbone frozen, yet delivers the accuracy above. In contrast, UNet-ResNet34 has ~24.4M trainable params (~93 MB) and still lags in IoU/Dice/OA. Despite requiring longer wall-clock time to train the FM decoder (ViT forward is heavy even when frozen), inference is dramatically lighter: your measurements show sub-millisecond/image for Copernicus-FM vs ~223–230 ms/image for UNet on the same GPU. For resource-constrained agencies, this translates to cheaper storage, faster on-device prediction, and simpler field deployment.

IoU difference between Copernicus-FM and UNet across fractions:
10%: +0.355, **20%:** +0.408, **30%:** +0.347, **50%:** +0.390, and **100%:** +0.376

Together, these two properties—fewer labels and lower deployment cost—directly address affordability and accessibility in regions with limited annotation capacity, limited connectivity, or small compute budgets.

RQ2: How effectively can Copernicus-FM perform when faced with a dataset beyond its pretraining horizon?

We deliberately evaluate on a non-benchmark, application-driven "water body" dataset whose curation differs from the Copernicus benchmark and lacks the rich metadata typically leveraged by geospatial FMs. Under this distribution shift, the frozen Copernicus-FM encoder still performs strongly: with full supervision (100%), we obtain Test IoU ≈ 0.75 / Dice ≈ 0.86, while 10% labels already yield ≈ 0.68 / 0.81, respectively. Crucially, this is achieved without unfreezing the backbone, while the fully fine-tuned UNet baseline remains far behind at every label fraction. This is the expected signature of a robust, broadly transferable FM: high absolute accuracy, small generalization gaps, and early saturation with limited labels.

Conclusion

This study demonstrates that a frozen Copernicus-FM encoder, paired with a lightweight decoder, achieves state-of-the-art performance in surface-water mapping while operating under strict label budgets and deployment constraints. Across varying levels of supervision (from 10% to 100%), Copernicus-FM consistently outperforms a strong UNet-ResNet34 baseline by approximately 0.35 to 0.41 in Intersection over Union (IoU), all while utilizing a trainable head that is roughly seven times smaller and enabling sub-millisecond inference. This showcases both label efficiency and cost-effectiveness in operational settings. Notably, these improvements are evident even on a non-benchmark, application-driven dataset that lacks metadata, suggesting the model's strong ability to transfer knowledge beyond its pretraining scope.
Limits and next steps: The dataset's masks originate from index-based procedures refined by manual checks; residual label noise likely caps the final IoU/Dice and explains mild non-monotonicity at 50%. Future work should (i) incorporate noise-robust losses and calibration metrics, (ii) expand to multi-sensor inputs (e.g., S1 SAR for cloud-robustness) without sacrificing the lightweight head, and (iii) add uncertainty quantification to support risk-aware decision-making during floods.